Stationary Random Processes

Lecture 7

October 28, 30, and November 4 and 6, 2025

Outline

1. Outline
2. Stationary Random Processes
3. Linear System with Random Process Input

Strict and Wide-Sense Stationarity

- Stationarity refers to time invariance of some, or all, of the statistics of a random process, such as mean, autocorrelation, nth-order distribution
- We define two types of stationarity: strict sense (SSS) and wide sense (WSS)
- A random process $X(t)$ (or X_n) is said to be SSS if all its finite order distributions are time invariant, i.e., the joint cdfs (pdfs, pmfs) of

$$X(t_1), X(t_2), \cdots, X(t_k) \text{ and } X(t_1+\tau), X(t_2+\tau), \cdots, X(t_k+\tau)$$

  are the same for all k, all $t_1, t_2, \cdots, t_k$, and all time shifts τ. For instance, for $k = 2$, we must have

$$f_{X(t_1),X(t_2)}(x_1, x_2) = f_{X(t_1+\tau),X(t_2+\tau)}(x_1, x_2) \text{ for every } -\infty < x_1, x_2 < \infty$$

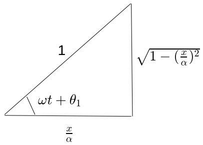
- So for an SSS process, the first-order distribution is independent of t, and the second-order distribution (the distribution of any two samples $X(t_1)$ and $X(t_2)$) depends only on $\tau = t_2 - t_1$: the joint distribution of $X(t_1)$ and $X(t_2)$ is the same as the joint distribution of $X(t_1 + (t - t_1)) = X(t)$ and $X(t_2 + (t - t_1)) = X(t + (t_2 - t_1))$

- Example: The random phase signal $X(t) = \alpha\cos(\omega t + \Theta)$, $-\infty < t < \infty$, where $\Theta \sim \text{Uniform}[0, 2\pi]$, and α and ω are constants, is SSS.
  - First-order pdf $f_{X(t)}(x)$: Fix x, then $\alpha\cos(\omega t + \theta) = x$ has two solutions in $[0, 2\pi]$ for $-\alpha \le x \le \alpha$:

$$\theta_1 = \cos^{-1}(x/\alpha) - \omega t\,, \quad \theta_2 = 2\pi - \theta_1$$

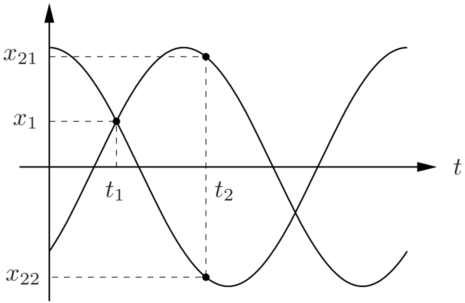
(where θ_1 is actually modulo π: principal value). We have

$$\frac{dx}{d\theta} = -\alpha\sin(\omega t + \theta) = -\alpha\sqrt{1-(x/\alpha)^2} \Rightarrow \left|\frac{dx}{d\theta}\right| = \alpha\sqrt{1-(x/\alpha)^2}$$



This leads to $f_{X(t)}(x) = \frac{1}{\alpha\pi\sqrt{1-(x/\alpha)^2}}$, $-\alpha < x < \alpha$ which is independent of t, and is therefore (first-order) stationary.

- $X(t) = \alpha \cos(\omega t + \Theta)$
  - To find the second-order pdf $f_{X(t_1),X(t_2)}(x_1, x_2)$: note that if we are given the value of $X(t)$ at one point, say t_1, there are (at most) two possible sample functions:



The second order pdf can thus be written as

$$\begin{aligned} f_{X(t_1),X(t_2)}(x_1, x_2) &= f_{X(t_1)}(x_1) f_{X(t_2)|X(t_1)}(x_2|x_1) \\ &= f_{X(t_1)}(x_1)\Big(\frac{1}{2}\delta(x_2 - x_{21}) + \frac{1}{2}\delta(x_2 - x_{22})\Big), \end{aligned}$$

which depends only on $t_2 - t_1$, and thus the second-order pdf is stationary.

- $X(t) = \alpha \cos(\omega t + \Theta)$
  - Now if we know that $X(t_1) = x_1$ and $X(t_2) = x_2$, the sample path is totally determined (except when $x_1 = x_2 = 0$, where two paths may be possible), and thus all n-th order pdfs are stationary. Thus, $X(t)$ is SSS.
- IID processes are SSS
- Random walk and Poisson counting processes are not SSS (they are not even first-order stationary)
  - Random Walk:

$$P(X_n = k) = \binom{n}{\frac{n+k}{2}} 2^{-n} \text{ for } -n \leq k \leq n \;\Rightarrow\; P(X_n = k) \neq P(X_{n+n_\tau} = k)$$

  - Poisson counting process:

$$P(N(t) = k) = \frac{[\lambda t]^k}{k!} e^{-\lambda t} u(k) \;\Rightarrow\; P(N(t) = k) \neq P(N(t+\tau) = k)$$

Wide-Sense Stationary Random Processes

- A random process $X(t)$ (or X_n) is said to be wide-sense stationary if its mean and autocorrelation functions are time invariant, i.e.,
  - $E\{X(t)\} = \mu$, $-\infty < t < \infty$, independent of t
  - $R_X(t_1, t_2)$ is a function only of the time difference $t_2 - t_1$. That is, let $t_2 = t + \tau$ and $t_1 = t$. Then $R_X(t, t + \tau)$ is independent of t.
  - $E\{X^2(t)\} < \infty$ (a technical condition).
- Since $R_X(t_1, t_2) = R_X(t_2, t_1)$, for any wide sense stationary process $X(t)$, $R_X(t_1, t_2)$ is a function only of $|t_2 - t_1|$
- Clearly SSS $\Rightarrow$ WSS. The converse is not necessarily true

- Example: Let (for $-\infty < t < \infty$)

$$X(t) = \begin{cases} \sin(t) & \text{with probability } \frac{1}{4} \\ -\sin(t) & \text{with probability } \frac{1}{4} \\ \cos(t) & \text{with probability } \frac{1}{4} \\ -\cos(t) & \text{with probability } \frac{1}{4} \end{cases}$$

  - $E\{X(t)\} = 0$ and $R_X(t_1, t_2) = \frac{1}{2}\cos(t_2 - t_1)$. Therefore, $X(t)$ is WSS.
  - But $X(0)$ and $X(\pi/4)$ do not have the same pmf (different ranges), so the first-order pmf is not stationary, and the process is not SSS
- Random walk and Poisson counting processes are not WSS
  - Random Walk: $E\{X_n\} = 0$ and $R_X(n_1, n_2) = \min(n_1, n_2)$; latter depends on n_1 and n_2 (not time-invariant: function only of $|n_2 - n_1|$), therefore, random walk is not WSS.
  - Poisson counting process: $E\{X(t)\} = \lambda t$ ($t \geq 0$): not time-invariant, therefore, Poisson counting process is not WSS.

Autocorrelation Function of WSS Processes

- Let $X(t)$ be a WSS process. Then since $R_X(t, t+\tau)$ is independent of t, we re-label it (or $R_X(t_1, t_2)$) as $R_X(\tau)$ with $\tau = t_2 - t_1$.
  1. $R_X(\tau)$ is real-valued and even function of τ, i.e., $R_X(\tau) = R_X(-\tau)$, $-\infty < \tau < \infty$
  2. $|R_X(\tau)| \leq R_X(0) = E\{X^2(t)\}$, the "average power" of (voltage waveform) $X(t)$
     To see this, consider

$$\begin{aligned} 0 \leq & E\{(X(t) \pm X(t+\tau))^2\} \\ = & \underbrace{E\{X^2(t)\}}_{R_X(0)} + \underbrace{E\{X^2(t+\tau)\}}_{R_X(0)} \pm 2\,\underbrace{E\{X(t)X(t+\tau)\}}_{R_X(\tau)} \\ = & 2\,[R_X(0) \pm R_X(\tau)] \Rightarrow -R_X(0) \leq R_X(\tau) \leq R_X(0) \;\Rightarrow\; |R_X(\tau)| \leq R_X(0) \end{aligned}$$

  3. If $R_X(T) = R_X(0)$ for some $T \neq 0$, then $R_X(\tau)$ is periodic with period T and so is $X(t)$ (with probability one). That is,

$$R_X(\tau) = R_X(\tau+T)\ \forall \tau \quad \text{and} \quad X(t) = X(t+T)\ \forall t \text{ (with probability one)}$$

- Example: The autocorrelation function for the periodic signal with random phase $X(t) = \alpha \cos(\omega t + \Theta)$ is $R_X(\tau) = \frac{\alpha^2}{2} \cos(\omega\tau)$, which is also periodic.
- The above properties of $R_X(\tau)$ are necessary but not sufficient for a function to qualify as an autocorrelation function for a WSS process
- The necessary and sufficient conditions for a function to be an autocorrelation function for a WSS process is that it be real, even, and nonnegative definite.

  By nonnegative definite we mean that for any n, any $t_1, t_2, \cdots, t_n$ and any real vector $\boldsymbol{a} = [a_1, a_2, \cdots, a_n]^\top$,

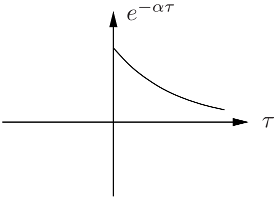
  $$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j R_X(t_i - t_j) \geq 0$$

  To see why this is necessary, recall that the correlation matrix for a random vector must be nonnegative definite, so if we take a set of n samples (at times $t_1, t_2, \cdots, t_n$) from the WSS random process $X(t)$, their correlation matrix must be nonnegative definite.

  The condition is sufficient since such an $R_X(\tau)$ can specify a zero mean stationary Gaussian random process (not discussed yet !)

Which Functions Can Be an $R_X(\tau)$?

1.



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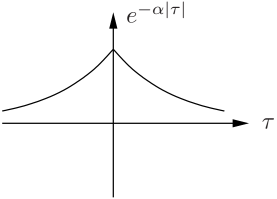

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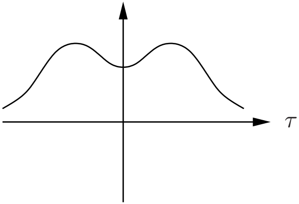

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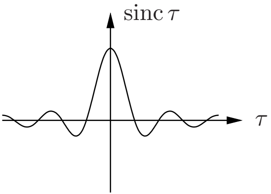
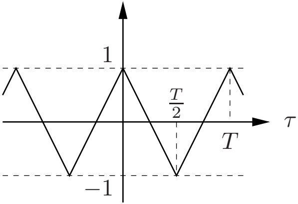


Which Functions Can Be an $R_X(\tau)$?

5.



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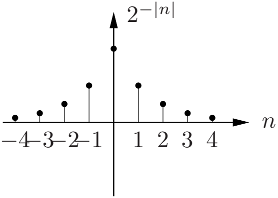

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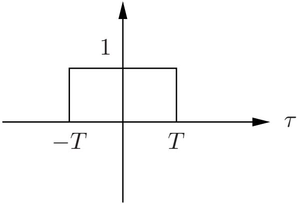

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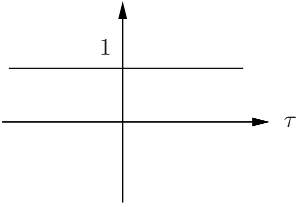


Power Spectral Density

- The power spectral density (psd) of a WSS random process $X(t)$ is the Fourier transform of $R_X(\tau)$:

$$R_X(\tau) \Leftrightarrow S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi\tau f}\, d\tau\,, -\infty < f < \infty$$

- For a discrete-time WSS random process X_n, the psd is the discrete-time Fourier transform (DTFT) of $R_X(n$:

$$R_X(n) \Leftrightarrow S_X(f) = \sum_{n=-\infty}^{\infty} R_X(n) e^{-j2\pi nf}$$

  Note that the DTFT $S_X(f) = S_X(f+m)$, $m = \pm 1, \pm 2, \cdots$, (i.e., periodic in f with period one), therefore, it is enough to specify it for $|f| \le \frac{1}{2}$.

- $R_X(\tau)$ (or $R_X(n)$) can be recovered from $S_X(f)$ by taking the inverse Fourier transform or inverse DTFT:

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi\tau f}\, df\,, \quad R_X(n) = \int_{-\frac{1}{2}}^{\frac{1}{2}} S_X(f) e^{j2\pi nf}\, df$$

Properties of the Power Spectral Density

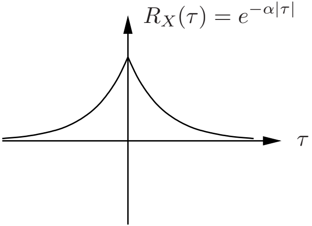
1. $S_X(f)$ is real and even, since the Fourier transform of the real and even function $R_X(\tau)$ is real and even
2. $\int_{-\infty}^{\infty} S_X(f)\, df = R_X(0) = E\{X^2(t)\}$, the average power of$X(t)$, i.e., the area under $S_X(f)$ is the average power
3. $S_X(f)$ is the average-power density, i.e., the average power of $X(t)$ in the frequency band $[f_1, f_2]$ is (this will be shown later when we consider linear filtering of $X(t)$)

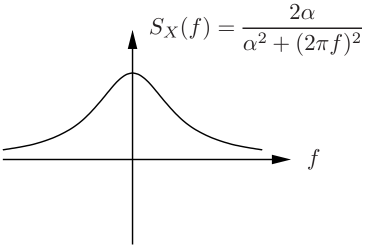
$$\int_{-f_2}^{-f_1} S_X(f)\, df + \int_{f_1}^{f_2} S_X(f)\, df = 2\int_{f_1}^{f_2} S_X(f)\, df\, .$$

- From property 3, it follows that $S_X(f) \geq 0$. Why? (Pick $f_1 = f_0 - \Delta$, $f_2 = f_0 + \Delta$, $\Delta \to 0$. If $S_X(f)$ is "smooth," $\int_{f_1}^{f_2} S_X(f)\, df \approx 2\Delta S_X(f_0) \geq 0 \Rightarrow S_X(f_0) \geq 0$. This is true for any f_0)
- In general, a function $S(f)$ is a psd (for some WSS $X(t)$) if and only if it is real, even, nonnegative, and $\int_{-\infty}^{\infty} S(f)\, df < \infty$.

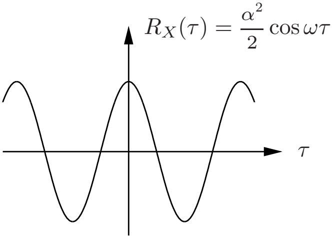
Examples

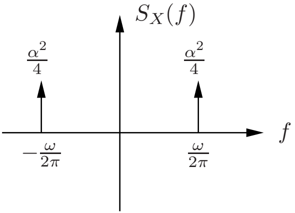
1.


$R_X(\tau) = e^{-\alpha|\tau|}$
τ

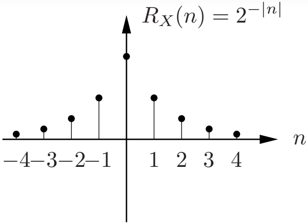

$S_X(f) = \dfrac{2\alpha}{\alpha^2 + (2\pi f)^2}$
f

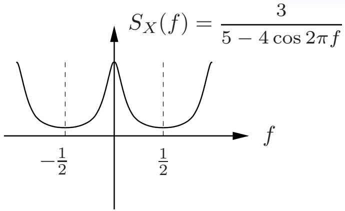
2.


$R_X(\tau) = \dfrac{\alpha^2}{2} \cos \omega\tau$
τ

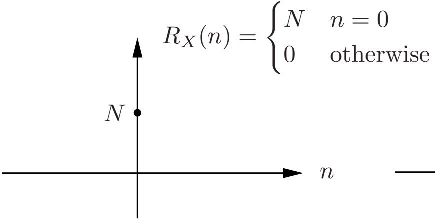

$S_X(f)$
$\frac{\alpha^2}{4}$
$\frac{\alpha^2}{4}$
f
$-\frac{\omega}{2\pi}$
$\frac{\omega}{2\pi}$

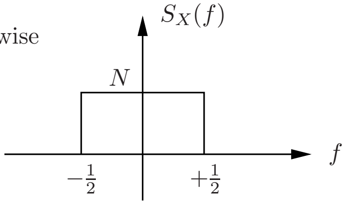
3.



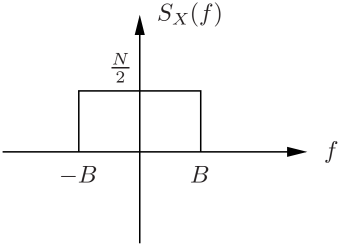


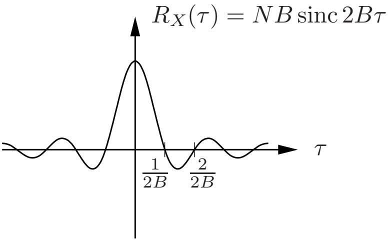
4 Discrete-time white noise process: $X_1, X_2, \cdots, X_n, \cdots$ zero-mean, uncorrelated, with average power N





5 Bandlimited white noise process: WSS zero mean process $X(t)$ with





For any t, the samples $X(t \pm \frac{n}{2B})$ for $n = 0, \pm 1, \pm 2, \cdots$ are uncorrelated

6 White noise process: If we let $B \to \infty$ in the previous example, we obtain a white noise process, which has

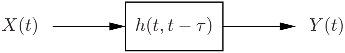
$$S_X(f) = \frac{N}{2}\,, -\infty < f < \infty \text{ and } R_X(\tau) = \frac{N}{2}\delta(\tau)\,, -\infty < \tau < \infty$$

Remarks on white noise

- For a white noise process, all samples are uncorrelated
- The process is not physically realizable, since it has infinite power: $E\{X^2(t)\} \to \infty$
- However, it plays a similar role in random processes to delta function in linear systems
- Thermal noise is well modeled as white Gaussian noise, since it has very flat psd over a very wide band (GHz)

Linear System with Random Process Input

- Consider a linear system with (time-varying) impulse response $h(t, t-\tau)$ driven by a random process input $X(t)$



- The output of the system is

$$Y(t) = \int_{-\infty}^{\infty} h(t, t-\tau) X(\tau)\, d\tau$$

- We wish to specify the output random process $Y(t)$. It is difficult to obtain a complete specification of the output process in general (except for Gaussian processes – a topic to be discussed later)
- We focus on finding the mean and autocorrelation functions of $Y(t)$ in terms of the mean and autocorrelation functions of the input process $X(t)$ and the impulse response of the system $h(t, t-\tau)$.

- We are also interested in finding the crosscorrelation function between $X(t)$ and $Y(t)$ defined as

$$R_{XY}(t_1, t_2) = E\{X(t_1)Y(t_2)\} = R_{YX}(t_2, t_1)$$

Note that unlike $R_{XX}(t_1, t_2)$, $R_{XY}(t_1, t_2)$ is not necessarily symmetric in t_1 and t_2.

- To find the mean of $Y(t)$, consider

$$E\{Y(t)\} = E\{\int_{-\infty}^{\infty} h(t, t-\tau)X(\tau)\, d\tau\} = \int_{-\infty}^{\infty} h(t, t-\tau)E\{X(\tau)\}\, d\tau$$

- The crosscorrelation function between $Y(t)$ and $X(t)$ is

$$\begin{aligned} R_{YX}(t_1, t_2) &= E\{Y(t_1)X(t_2)\} \\ &= E\{\int_{-\infty}^{\infty} h(t_1, t_1 - \tau)X(\tau)X(t_2)\, d\tau\} \\ &= \int_{-\infty}^{\infty} h(t_1, t_1 - \tau)E\{X(\tau)X(t_2)\}\, d\tau \\ &= \int_{-\infty}^{\infty} h(t_1, t_1 - \tau)R_X(\tau, t_2)\, d\tau \end{aligned}$$

- The autocorrelation function of $Y(t)$ is

$$
\begin{aligned}
R_Y(t_1, t_2) &= E\{Y(t_1)Y(t_2)\} = E\{Y(t_2)Y(t_1)\} \\
&= E\{\int_{-\infty}^{\infty} h(t_2, t_2 - \tau)X(\tau)Y(t_1)\, d\tau\} \\
&= \int_{-\infty}^{\infty} h(t_2, t_2 - \tau)\, \underbrace{E\{X(\tau)Y(t_1)\}}_{R_{YX}(t_1,\tau)}\, d\tau \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(t_2, t_2 - \tau_2)h(t_1, t_1 - \tau_1)R_X(\tau_1, \tau_2)\, d\tau_1\, d\tau_2
\end{aligned}
$$

The average power is $E\{Y^2(t)\} = R_Y(t, t)$

- Example (Integrator): Let $X(t)$ be a white noise process with autocorrelation function $R_X(\tau) = (N/2)\delta(\tau)$ and let the linear system be an ideal integrator, i.e.,

$$Y(t) = \int_0^t X(\tau)\, d\tau$$

Find the mean and autocorrelation functions and the average power of the integrator output $Y(t)$ for $t \geq 0$

- This example is motivated by several applications:
  - Noise in an image sensor pixel: the white noise models the photodetector shot noise, which is integrated with the signal over a capacitor before sampling
  - Noise in a voltage controlled oscillator (for phase locked loops)
- Solution: The mean is

$$E\{Y(t)\} = \int_0^t E\{X(\tau)\}\,d\tau = 0$$

To obtain the autocorrelation function and average power for this case, we can specialize the previous results to

$$\begin{aligned} R_{YX}(t_1, t_2) = E\{Y(t_1)X(t_2)\} &= \int_0^{t_1} R_X(\tau, t_2)\,d\tau \\ &= \int_0^{t_1} (N/2)\delta(\tau - t_2)\,d\tau \\ &= \begin{cases} \frac{N}{2} & \text{if } t_2 \le t_1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

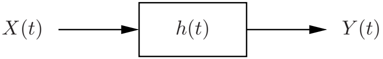
- $$R_Y(t_1, t_2) = \int_0^{t_2} E\{Y(t_1)X(\tau)\}\, d\tau = \int_0^{t_2} R_{YX}(t_1, \tau)\, d\tau$$
$$= \begin{cases} \int_0^{t_2} \frac{N}{2}\, d\tau = \frac{N}{2} t_2 & \text{if } t_2 \leq t_1 \\ \int_0^{t_1} \frac{N}{2}\, d\tau = \frac{N}{2} t_1 & \text{otherwise} \end{cases}$$
$$= \frac{N}{2} \min(t_1, t_2)$$

- The average power is

$$E\{Y^2(t)\} = R_Y(t, t) = \frac{N}{2} t\,, \quad t \geq 0$$

LTI System with WSS Process Input

- Consider a linear time invariant (LTI) system with real impulse response $h(t)$ and transfer function $H(f)$ (Fourier transform of $h(t)$), driven by WSS process $X(t)$, $-\infty < t < \infty$.



- We want to characterize its output
  $Y(t) = h(t) * X(t) = \int_{-\infty}^{\infty} h(t-\tau)X(\tau)\, d\tau$
- It turns out (not surprisingly) that if the system is stable, i.e., if $|\int_{-\infty}^{\infty} h(t)\, dt| = |H(0)| < \infty$, then $X(t)$ and $Y(t)$ are jointly WSS, which means that:
  - $X(t)$ and $Y(t)$ are WSS
  - Their crosscorrelation function $R_{XY}(t_1, t_2)$ is time invariant, i.e.,

$$R_{XY}(t_1, t_2) = E\{X(t_1)Y(t_2)\} = R_{XY}(t_1 + \tau, t_2 + \tau) \text{ for all } \tau$$

- Relabel $R_{XY}(t_1, t_2)$ for jointly WSS $X(t)$, $Y(t)$ as $R_{XY}(\tau)$ where $\tau = t_1 - t_2$:

$$R_{XY}(\tau) = R_{XY}(t_2 + \tau, t_2) = R_{XY}(t_2 + (t_1 - t_2), t_2) = R_{XY}(t_1, t_2)$$

  Note that $R_{XY}(\tau)$ is not necessarily even. However, $R_{XY}(\tau) = R_{YX}(-\tau)$.

- Example: Let $\Theta \sim \text{Uniform}[0, 2\pi]$. Consider two processes

$$X(t) = \alpha \cos(\omega t + \Theta) \text{ and } Y(t) = \alpha \sin(\omega t + \Theta)$$

  These processes are jointly WSS, since each is WSS (in fact SSS) and

$$\begin{aligned}
R_{XY}(t_1, t_2) &= E\{\alpha^2 \underbrace{\cos(\omega t_1 + \Theta)\sin(\omega t_2 + \Theta)}_{0.5\sin(\omega(t_1+t_2)+2\Theta)+0.5\sin(\omega(t_2-t_1))}\} \\
&= \frac{\alpha^2}{4\pi}\int_0^{2\pi} \left[\sin(\omega(t_1 + t_2) + 2\theta) + \sin(\omega(t_2 - t_1))\right] d\theta \\
&= \frac{\alpha^2}{4\pi}\left[0 + 2\pi \sin(\omega(t_2 - t_1))\right] \\
&= \frac{\alpha^2}{2}\sin(\omega(t_2 - t_1)) \to -\frac{\alpha^2}{2}\sin(\omega\tau)
\end{aligned}$$

- We define the cross power spectral density for jointly WSS processes $X(t)$, $Y(t)$ as

$$R_{XY}(\tau) \Leftrightarrow S_{XY}(f) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j2\pi\tau f}\, d\tau$$

- Example: Let $Y(t) = X(t) + Z(t)$, where $X(t)$ and $Z(t)$ are zero mean uncorrelated WSS processes. Show that $Y(t)$ and $X(t)$ are jointly WSS, and find $R_{XY}(\tau)$ (in terms of R_X and R_Z) and $S_{XY}(f)$ (in terms of S_X and S_Z)

  Solution: First, we show that $Y(t)$ is WSS. It is zero mean and

$$\begin{aligned} R_{YY}(t_1, t_2) &= E\{[X(t_1) + Z(t_1)][X(t_2) + Z(t_2)]\} \\ &= E\{X(t_1)X(t_2)\} + E\{Z(t_1)Z(t_2)\} + 0 + 0 \\ &= R_X(\tau) + R_Z(\tau) = R_Y(\tau) \text{ where } \tau = t_1 - t_2 \end{aligned}$$

  Take the Fourier transform on both sides: $S_Y(f) = S_X(f) + S_Z(f)$.

- To show that $Y(t)$ and $X(t)$ are jointly WSS, we need to show that their crosscorrelation function is time invariant:

$$\begin{aligned} R_{XY}(t_1, t_2) &= E\{X(t_1)[X(t_2) + Z(t_2)]\} \\ &= E\{X(t_1)X(t_2)\} + E\{X(t_1)Z(t_2)\} \\ &= R_X(t_1, t_2) + E\{X(t_1)\}E\{Z(t_2)\} = R_X(t_1, t_2) \\ &= R_X(\tau) \text{ where } \tau = t_1 - t_2 \\ &= R_{XY}(\tau) \end{aligned}$$

Finally, take the Fourier transform on both sides:

$$S_{XY}(f) = S_X(f)$$

Output Mean, Autocorrelation, and PSD

Theorem: Let $X(t)$, $-\infty < t < \infty$, be a WSS process input to a stable LTI system with real impulse response $h(t)$ and transfer function $H(f)$. Then the input $X(t)$ and output $Y(t)$ are jointly WSS with

1. $E\{Y(t)\} = H(0)E\{X(t)\}$
2. $R_{YX}(\tau) = h(\tau) * R_X(\tau) = \int_{-\infty}^{\infty} h(u) R_X(\tau - u)\, du$
3. $R_Y(\tau) = h(\tau) * R_X(\tau) * h(-\tau)$

$$R_X(\tau) \longrightarrow \boxed{h(\tau)} \xrightarrow{R_{YX}(\tau)} \boxed{h(-\tau)} \longrightarrow R_Y(\tau)$$

4. $S_{YX}(f) = H(f)S_X(f)$
5. $S_Y(f) = H(f)S_X(f)H(-f) = |H(f)|^2 S_X(f)$

$$S_X(f) \longrightarrow \boxed{H(f)} \xrightarrow{S_{YX}(f)} \boxed{H(-f)} \longrightarrow S_Y(f)$$

Proof: Note that here the LTI system is in steady state.

1. Since $X(t)$ is WSS, $E\{X(t)\} = \mu_X$ for any t.

$$
\begin{aligned}
E\{Y(t)\} &= E\{\int_{-\infty}^{\infty} h(\tau)X(t-\tau)\,d\tau\} \\
&= \int_{-\infty}^{\infty} h(\tau)E\{X(t-\tau)\}\,d\tau \\
&= \mu_X \int_{-\infty}^{\infty} h(\tau)\,d\tau = \mu_X H(0) = H(0)E\{X(t)\}
\end{aligned}
$$

2. To find the crosscorrelation function between $Y(t)$ and $X(t)$, consider

$$
\begin{aligned}
R_{YX}(\tau) &= E\{Y(t+\tau)X(t)\} = E\{\int_{-\infty}^{\infty} h(u)X(t+\tau-u)X(t)\,du\} \\
&= \int_{-\infty}^{\infty} h(u)R_X(\tau-u)\}\,du \\
&= h(\tau) * R_X(\tau)
\end{aligned}
$$

3. To find the autocorrelation function of $Y(t)$, consider

$$
\begin{aligned}
R_Y(\tau) &= E\{Y(t+\tau)Y(t)\} \\
&= E\{Y(t+\tau)\int_{-\infty}^{\infty} h(u)X(t-u)\,du\} \\
&= \int_{-\infty}^{\infty} h(u)R_{YX}(\tau+u)\}\,du \\
&\overset{v=-u}{=} \int_{-\infty}^{\infty} h(-v)R_{YX}(\tau-v)\}\,dv \\
&= h(-\tau) * R_{YX}(\tau)
\end{aligned}
$$

4. Follows by taking the Fourier transform of $R_{YX}(\tau)$
5. Follows by taking the Fourier transform of $R_Y(\tau)$

Remark: For a discrete-time WSS process $X(n)$ (or X_n) and a stable LTI system $h(n)$, $X(n)$ and the output process $Y(n)$ are jointly WSS and we can similarly find $R_Y(n)$, $\cdots$

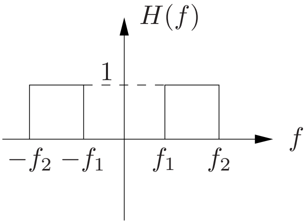
$S_X(f)$ is the Power Spectral Density

- We can use the above results to show that $S_X(f)$ is indeed the power spectral density of $X(t)$; i.e., the average power in any frequency band $f_1, f_2]$ is

$$2\int_{f_1}^{f_2} S_X(f)\, df$$

- To show this we pass $X(t)$ through an ideal band-pass filter

$$X(t) \longrightarrow \boxed{h(t)} \longrightarrow Y(t)$$



- Now the average power of $X(t)$ in the band $f_1, f_2]$ is

$$
\begin{aligned}
E\{Y^2(t)\} &= \int_{-\infty}^{\infty} S_Y(f)\,df = \int_{-\infty}^{\infty} |H(f)|^2 S_X(f)\,df \\
&= \int_{-f_2}^{-f_1} S_X(f)\,df + \int_{f_1}^{f_2} S_X(f)\,df \\
&= 2\int_{f_1}^{f_2} S_X(f)\,df
\end{aligned}
$$

- This also shows that $S_X(f) \geq 0$ for all f.

  Pick $f_1 = f_0 - \Delta$, $f_2 = f_0 + \Delta$, $\Delta \to 0$. If $S_X(f)$ is "smooth,"
  $\int_{f_1}^{f_2} S_X(f)\,df \approx 2\Delta S_X(f_0) \geq 0 \Rightarrow S_X(f_0) \geq 0$. This is true for any f_0